

RUFUS BLESSING

Generative AI Engineer | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Generative AI Engineer and Machine Learning Engineer with 3 years of experience designing and deploying scalable AI-powered applications, intelligent automation systems, and Retrieval-Augmented Generation (RAG) solutions. Skilled in Python, FastAPI, React/Next.js, Prompt Engineering, LLM Fine-Tuning, Hugging Face, TensorFlow, Scikit-learn, PostgreSQL, PGVector, Docker, and MLflow. Experienced in building production-ready enterprise AI assistants, vector search systems, and cloud-based workflows across AWS, Azure, and GCP. Strong background in predictive modeling, feature engineering, and end-to-end machine learning pipelines, with a proven track record of delivering secure, scalable, and business-impactful AI solutions.

TECHNICAL SKILLS

Programming Languages: Python (Intermediate, HackerRank Certified), SQL

AI & ML Frameworks: Scikit-learn, TensorFlow, Hugging Face, PyTorch (learning)

Generative AI: Prompt Engineering, RAG Pipelines, LLM Fine-Tuning, Embeddings, Vector Search

Machine Learning: Regression, Classification, Clustering, Model Evaluation (MSE, R-squared, Accuracy)

Data Science: Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering

Data Visualization: Matplotlib, Seaborn, Plotly; exposure to Tableau and Power BI

Databases: PostgreSQL, MySQL; exposure to non-relational database concepts

Cloud & MLOps: MQTT, Docker, MLflow, AWS, Azure, GCP

Version Control: Git, GitHub

PROFESSIONAL EXPERIENCE

Generative AI Engineer — Honeycone Corp, Chennai

2023 – Present

- Led development of LLM-powered applications, including RAG-based assistants and fine-tuned transformer models.
- Designed and deployed scalable Generative AI workflows integrating Hugging Face models with vector search databases.
- Collaborated with cross-functional teams to deliver AI-driven solutions for knowledge retrieval, summarization, and automation.
- Implemented containerized deployments using FastAPI and Docker, with monitoring via MLflow.
- Mentored junior engineers on prompt engineering and model evaluation best practices.

PROJECTS

Enterprise AI Knowledge Assistant

React/Next.js, FastAPI, PostgreSQL + PGVector, OpenAI/Azure/Claude, Docker, Prometheus/Grafana/Loki

- Architected and delivered a production-ready Enterprise AI Knowledge Assistant for organizational knowledge management.
- Built a React/Next.js UI with chat interface and admin dashboard, secured via HTTPS and JWT authentication.
- Developed a FastAPI gateway handling authentication, user management, chat APIs, upload APIs, search APIs, and admin APIs.

Modular services implemented:

- Document Service: upload processing, parsing, OCR, chunking.
- RAG Service: semantic search, embeddings, retrieval, LLM integration.
- User Service: authentication, RBAC, JWT, audit logs.
- Integrated PostgreSQL with PGVector for semantic search and retrieval.
- Connected to OpenAI, Azure OpenAI, Claude, and local LLMs for flexible inference.
- Established monitoring stack with Prometheus, Grafana, and Loki for observability and logging.

- Delivered a scalable, secure, enterprise-grade AI assistant that improved knowledge access and productivity across teams.

Edge Predictive Maintenance System

Python, FastAPI, Scikit-learn, MQTT, PostgreSQL, Streamlit

- Developed a real-time predictive maintenance system to monitor machine health and predict failures.
- Built RESTful APIs with FastAPI to serve machine learning predictions.
- Implemented MQTT-based communication for low-latency sensor data transfer.
- Designed interactive Streamlit dashboards for visualization and alerts.

EDUCATION

MSc Computer Science (79%) — Nandanam Arts Government College, Chennai (2023)

Bachelor of Computer Applications (77%) — Mar Gregorios College, Chennai (2021)

HSC — Madras Christian College Higher Secondary School, Chennai (2018)

CERTIFICATIONS

Python — HackerRank Certified